

Work through the following exercises while the TA circulates. When you have completed the exercises, spend the rest of the laboratory working on the homework assignment.

Warm up.

Compute the following integrals.

1. $\int e^{-x} dx$.
2. $\int e^{6x} dx$.
3. $\int \sin(-3x) dx$.
4. $\int \frac{e^x}{e^x+1} dx$.
5. $\int x^3 \sqrt{x^2+1} dx$.
6. $\int \frac{\cos(x)}{\sin(x)^2} dx$.

Exercise 1.

Compute the integral $\int_1^2 \frac{\ln(x)^2}{x}$ in two ways.

1. Change the limits of integration when using a substitution (namely using the change of variables formula for definite integrals).
2. Find the general anti-derivative and then evaluate the limits of integration.

Exercise 2.

A function $f(x)$ is said to be *even* when $f(-x) = f(x)$ for all x in the domain of $f(x)$.

1. Sketch the graph of a continuous even function $f(x)$.
2. Explain why $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ for the even function $f(x)$ you sketched.
3. Recall that for continuous functions $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$. Use the substitution $u = -x$ on the first summand to prove that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ for every continuous even function $f(x)$.
4. The function $g(x) = \frac{1}{x^2}$ is even, but it is not continuous for $-1 \leq x \leq 1$. The integral of a function $h(x)$ continuous on $a \leq x < b$ and discontinuous at $x = b$ is $\int_a^b h(x) = \lim_{R \rightarrow b^-} \int_a^R h(x)$. The integral of a function $h(x)$ continuous on $a < x \leq b$ and discontinuous at $x = a$ is $\int_a^b h(x) = \lim_{R \rightarrow a^+} \int_R^b h(x)$. Compute $\int_{-a}^0 g(x) dx$, compute $\int_0^a g(x) dx$, and explain why you think the equalities $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ and $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$ are reasonable or not.